

DATA STRUCTURE

LAB PROGRAMS

24. Kruskal's Algorithm

```
#include <stdio.h>

struct Edge {
    int u,v,w;
};

int parent[10];

int find(int i) {
    while(parent[i])
        i=parent[i];
    return i;
}

int unionSet(int i,int j) {
    if(i!=j) {
        parent[j]=i;
        return 1;
    }
    return 0;
}

int main() {
    struct Edge edge[5] = {
        {0,1,10},
        {0,2,6},
        {0,3,5},
        {1,3,15},
        {2,3,4}
    }
```

```

};

int n=4,count=0,cost=0;

printf("Edges in MST:\n");

while(count < n-1) {

    int min=999,a=-1,b=-1,index=-1;

    for(int i=0;i<5;i++) {

        if(edge[i].w < min) {

            min=edge[i].w;

            a=edge[i].u;

            b=edge[i].v;

            index=i;

        }

    }

    int u=find(a);

    int v=find(b);

    if(unionSet(u,v)) {

        printf("%d - %d : %d\n",a,b,min);

        cost += min;

        count++;

    }

    edge[index].w = 999;

}

printf("Total Cost = %d",cost);

return 0;

}

```

Output

Edges in MST:

2 - 3 : 4

0 - 3 : 5

0 - 2 : 6

Total Cost = 15

=== Code Execution Successful ===